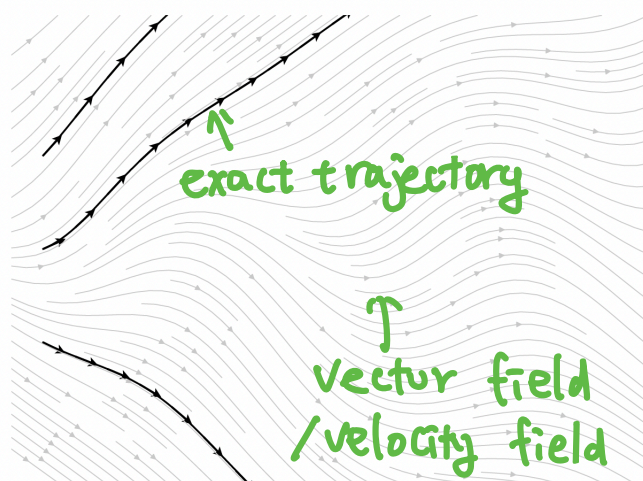
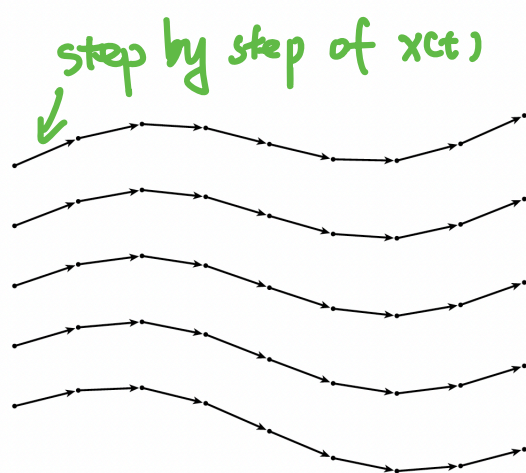


○ Differential Equations

- Density Evolution : Fokker-Planck
- Itô's Calculus and Girsanov's Theorem

Deterministic process can be described as an ODE:

$$\frac{dx(t)}{dt} = v(x(t), t) \quad \left\{ \begin{array}{l} x(t) : \mathbb{R}^D \text{ trajectory} \\ v(x(t)) : \mathbb{R}^D \times \mathbb{R} \rightarrow \mathbb{R}^D \text{ vector field} \end{array} \right.$$



Solving an ODE is like placing a particle in a flow field and observing where it goes over time!

○ IVP (Initial Value Problem) of ODEs

Local existence and uniqueness :

Theorem A.1.1: Local Existence and Uniqueness

Let $v(x, t)$ be a continuous function with respect to x and t in a domain $D \subseteq \mathbb{R}^D \times \mathbb{R}$. If v satisfies the Lipschitz condition with respect to x :

$$\|v(x_1, t) - v(x_2, t)\| \leq L \|x_1 - x_2\| \quad \forall (x_1, t), (x_2, t) \in D,$$

where $L > 0$ is a constant, then for every initial condition $x(t_0) = x_0$, there exists a unique solution $x(t)$ to Equation (A.1.1) defined on some interval $[t_0 - \delta, t_0 + \delta]$.

If we have the following two conditions:

- (i) **Local Lipschitz condition in $\mathbf{x}$:** There exists a function $\text{Lip}(t)$, integrable on $[0, T]$, such that for all $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^D$,

$$\|\mathbf{v}(\mathbf{x}_1, t) - \mathbf{v}(\mathbf{x}_2, t)\| \leq \text{Lip}(t) \|\mathbf{x}_1 - \mathbf{x}_2\|.$$

- (ii) **Linear growth condition:** There exists a function $M(t)$, integrable on $[0, T]$, such that for all $\mathbf{x} \in \mathbb{R}^D$,

$$\|\mathbf{v}(\mathbf{x}, t)\| \leq M(t)(1 + \|\mathbf{x}\|).$$

(*) $\Rightarrow$ IVP: $\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}(\mathbf{x}, t)$, $\mathbf{x}(0) = \mathbf{x}_0$, has unique solution.

Corollary A.1.1: Non-Intersection of Solutions

Consider two solutions $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ to the ODE

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}(\mathbf{x}(t), t), \quad t \in [0, T].$$

Suppose they have distinct initial values $\mathbf{x}_1(0) \neq \mathbf{x}_2(0)$. Then, these solutions do not intersect on $[0, T]$, i.e.,

$$\mathbf{x}_1(t) \neq \mathbf{x}_2(t) \quad \text{for all } t \in [0, T].$$

By using Gronwall's Inequality, we have:

Theorem 2.8. Suppose $f, g \in C(U, \mathbb{R}^n)$ and let f be locally Lipschitz continuous in the second argument, uniformly with respect to the first. If $x(t)$ and $y(t)$ are respective solutions of the IVPs

$$\begin{aligned} \dot{x} &= f(t, x) & \text{and} & & \dot{y} &= g(t, y) \\ x(t_0) &= x_0 & & & y(t_0) &= y_0 \end{aligned}, \quad (2.39)$$

then

$$|x(t) - y(t)| \leq |x_0 - y_0| e^{L|t-t_0|} + \frac{M}{L}(e^{L|t-t_0|} - 1), \quad (2.40)$$

where

$$L = \sup_{(t,x) \neq (t,y) \in V} \frac{|f(t,x) - f(t,y)|}{|x - y|}, \quad M = \sup_{(t,x) \in V} |f(t,x) - g(t,x)|, \quad (2.41)$$

with $V \subset U$ some set containing the graphs of $x(t)$ and $y(t)$.

When $f=g$. Denote the solution of (*) as $\phi(t, x_0)$
 $|\phi(t_0, x_0) - \phi(t_0, y_0)| \leq \|x_0 - y_0\| \cdot e^{L|t-t_0|}.$

Example of linear ODE:

$$\frac{dx(t)}{dt} = L(t) \cdot x(t) \Rightarrow x(t) = x(s) \cdot \exp\left(\int_s^t L(\tau) d\tau\right)$$

Example of semi-linear ODE:

$$\frac{dx(t)}{dt} = L(t) \cdot x(t) + N(x, t), \quad N: \mathbb{R}^D \times [0, T] \rightarrow \mathbb{R}^D \text{ nonlinear}$$

$$\Rightarrow x(t) = \exp\left(\int_s^t L(\tau) d\tau\right) \cdot x(s) + \int_s^t \exp\left(\int_s^t L(\tau) d\tau\right) \cdot N(x(\tau), \tau) d\tau$$

Numerical Solvers of ODEs

$$x(t) = x(0) + \int_0^t v(x(\tau), \tau) d\tau \Rightarrow \text{hard to calculate}$$

Main steps:

□ Discretization: $t: [0, T] \rightarrow t_0, t_1, t_2, \dots, t_n$

□ step size: $\Delta t_i = t_{i+1} - t_i = h$

□ approximation: based on h and numerical method

1. Time-stepping methods $\rightarrow$ $\begin{cases} \text{single-step: Euler, Heun, Runge-Kutta} \\ \text{multi-step: Adams-Bashforth, Adams-Moulton} \end{cases}$

2. Time-parallel methods

Euler's method: $x_{n+1} = x_n + h \cdot v(x_n, t_n)$

local error: $O(h^2)$, global error: $O(h)$

Heun's method: $\begin{cases} \text{predict: } x_{\text{pred}} = x_n + h \cdot v(x_n, t_n) \\ \text{correct: } x_{n+1} = x_n + \frac{h}{2} (v(x_n, t_n) + v(x_{\text{pred}}, t_n + h)) \end{cases}$

local error: $O(h^3)$, global error: $O(h^2)$

Runge-Kutta methods:

$$k_1 = v(x_n, t_n),$$

$$k_2 = v(x_n + \frac{h}{2}k_1, t_n + \frac{h}{2}),$$

$$k_3 = v(x_n + \frac{h}{2}k_2, t_n + \frac{h}{2}),$$

$$k_4 = v(x_n + hk_3, t_n + h),$$

$$x_{n+1} = x_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

local error: $O(h^5)$, global error: $O(h^4)$

Picard Iteration: $x^{(k+1)}(t) = x(0) + \int_0^t v(x^{(k)}(s), s) ds$

It converges slowly due to its dependence on $x(0)$.

Terminal Value problem: (usually in our denoising process)

$$\frac{dx(t)}{dt} = v(x(t), t), \quad x(T) = x_T \sim \text{prior}, \quad t \in [0, T]$$

Reparameterizing as an initial value problem:

Let $s := T - t$ (so, $t = T - s$), define the reparameterized path:

$$y(s) := x(T - s) \Rightarrow$$

$$\frac{dy(s)}{ds} = \frac{dx(T-s)}{ds} = x'(T-s) \cdot \frac{d(T-s)}{ds}$$

$$= -x'(T-s) = -v(x(T-s), T-s)$$

$$= -v(y(s), T-s)$$

$\Rightarrow$ We obtain an IVP problem:

$$\frac{dy(s)}{ds} = -v(y(s), T-s), \quad y(0) = x(T) = x_T$$

$y(T) = x(T-T) = x(0)$: the predicted image!

○ From ODE to SDE

From: $\frac{dx(t)}{dt} = f(x(t), t), x(0) = x_0$

Discretization perspective:

$$x_{t+\Delta t} = x_t + f(x_t, t) \cdot \Delta t, \Delta t \rightarrow 0$$

Note: $\frac{x_{t+\Delta t} - x_t}{\Delta t} \approx f'(x_t, t)$

ODE $\rightarrow$ SDE by introducing randomness:

$$x_{t+\Delta t} = x_t + f(x_t, t) \cdot \Delta t + g(t) \cdot \sqrt{\Delta t} \cdot \xi_t, \xi_t \sim N(0, I_D)$$

□ $g(t)$: diffusion coefficient

□ $\xi_t \sim N(0, I_D)$: i.i.d

□ $\sqrt{\Delta t}$ is the right scaling:

$$\Delta x_{\text{noise}} = g(t) \cdot \sqrt{\Delta t} \xi_t$$

$$\text{Var}[\Delta x_{\text{noise}}] = g(t)^2 \cdot \Delta t \cdot I$$

Variance of a single step Δt , $\rightarrow 0$ when $\Delta t \rightarrow 0$. $N = \frac{T}{\Delta t}$

total variance: $x_N = g^2 \cdot \sqrt{\Delta t}^2 \sum_{k=0}^{N-1} \xi_k, \text{Var}[x_N] \approx g^2 \cdot T$.

• If we do: $\Delta x_{\text{noise}} = g(t) \xi_t$

$$\text{Var}[\Delta x_{\text{noise}}] = g^2(t) \cdot I : N = \frac{T}{\Delta t} \text{ steps,}$$

total variance will blow up $\rightarrow \infty$.

• If we do: $\Delta x_{\text{noise}} = g(t) \cdot \Delta t \cdot \xi_t$

$$\text{Var}[\Delta x_{\text{noise}}] = g^2(t) \cdot \Delta t^2 \cdot I : N = \frac{T}{\Delta t} \text{ steps}$$

total variance will $g^2(t) \cdot \Delta t^2 \cdot \frac{T}{\Delta t} \sim \Delta t \rightarrow 0$.

The randomness disappears.

$$\text{SDE: } dx(t) = f(x(t), t)dt + g(t)I_D \cdot w(t) dt^{1/2}$$

$w(t) \in \mathbb{R}^D$: Wiener process (Brownian motion)

□ Initial state $w(0) = 0$ almost surely

□ Independent Increments: for $0 \leq s < t$, the increment $w(t) - w(s)$ is independent of the past

□ Gaussian Increments:

$$w(t) - w(s) \sim \mathcal{N}(0, (t-s)I_D)$$

□ Continuity: $t \mapsto w(s)$ almost surely continuous but nowhere differentiable.

$$\text{Notation: } \begin{cases} dw(t) := w(t+dt) - w(t) \\ dw(t) \sim \mathcal{N}(0, dt I_D) \end{cases}$$

SDE should be understood in its integral form:

$$\text{Itô form: } x(t) = x(0) + \int_0^t f(x(s), s) ds + \int_0^t g(s) dw(s)$$

$\int_0^t g(s) dw(s)$: Itô stochastic integral

$$\int_0^t g(s) dw(s) \approx \lim_{h \rightarrow 0} \sum_i g(t_i) \cdot (w(t_{i+1}) - w(t_i))$$

t_i : left endpoint of each subinterval $h_i = t_{i+1} - t_i$

$$\text{In ODE, } dx(t) = f(x(t), t) dt$$

$$x(t) = x(0) + \int_0^t f(x(\tau), \tau) d\tau \quad : \text{ by FTC}$$

$$\text{In SDE, } dx(t) = f(x(t), t) dt + g(x(t), t) dw(t)$$

$$x(t) = x(0) + \int_0^t f(x(\tau), \tau) d\tau + \underbrace{\int_0^t g(x(\tau), \tau) dw(\tau)}_{\substack{\uparrow \\ \text{nowhere differentiable}}}$$

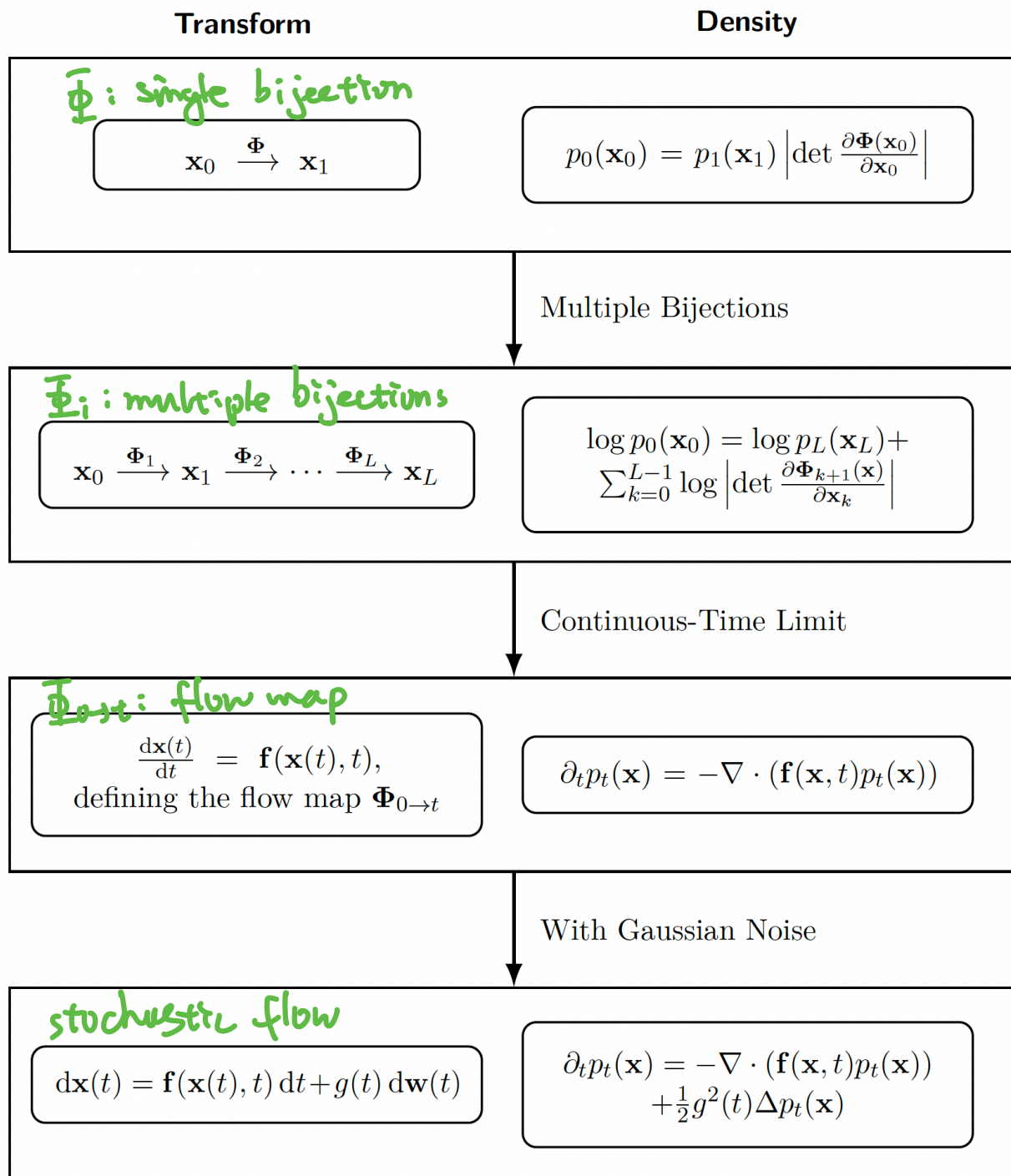
small Δt :

$$\text{ODE increment } \Delta x \approx f \cdot \Delta t$$

Wiener increment $\Delta w \sim \mathcal{N}(0, \Delta t)$, $|\Delta w| \sim \sqrt{\Delta t}$, $|\Delta w|^2 \sim \Delta t$: significant

- Differential Equations
- Density Evolution : Fokker-Planck
- Itô's Calculus and Girsanov's Theorem

$p_{\text{prior}} \sim p_{\text{data}}$: how probability densities evolve



$p(\mathbf{x}, t)$: how much probability mass in location $\mathbf{x}$ at time t .
 it doesn't disappear, but it only moves over time.

○ change-of-Variable Formula for Deterministic Maps

Study how their density evolves.

Single Update:

- Smooth and invertible mapping $\bar{\Phi} : \mathbb{R}^D \rightarrow \mathbb{R}^D$
- one unit of time $0 \rightarrow 1$, initial state x_0 , next state

$$x_1 = \bar{\Phi}(x_0)$$

- how density moves:

If $x_0 \sim p_0$, $0 \rightarrow 1$ with $\bar{\Phi}$: produces a new density

p_1 for x_1 , $x_1 \sim p_1$, p_1 is obtained by change-of-variable

$$p_1(x_1) = p_0(\bar{\Phi}^{-1}(x_1)) \cdot \left| \det \left[\frac{\partial \bar{\Phi}^{-1}}{\partial x_1} \right] \right|$$

□ A tiny region A around x_1 , preimage $\bar{\Phi}^{-1}(A)$ is a region around x_0 .

□ $\left| \det \left[\frac{\partial \bar{\Phi}^{-1}}{\partial x_1} \right] \right|$: volume stretch factor

Example: $\bar{\Phi} = A$ a invertible matrix, then

$$p_1(x_1) = p_0(A^{-1} \cdot x_1) \cdot |\det(A^{-1})|$$

$$x_0 \xrightarrow{\bar{\Phi}} x_1, \quad p_{x_0}(x_0) \xrightarrow{\bar{\Phi}} p_{x_1}(x_1)$$

Single Variable case: $y = \bar{\Phi}(x)$, $\bar{\Phi}$ is invertible

g : the function you are integrating (involve y)

$$\bar{g}(x) = \lim_{\Delta x \rightarrow 0} \frac{\bar{g}(x + \Delta x) - \bar{g}(x)}{\Delta x}$$

$$\Delta y := \bar{\Phi}(x+\Delta x) - \bar{\Phi}(x) \Rightarrow \Delta y \approx \bar{\Phi}'(x) \cdot \Delta x$$

= "infinitesimal" version of the approximation: $dy = \bar{\Phi}'(x) dx$

$$\int g(y) dy = \int g(\bar{\Phi}(x)) \cdot |\bar{\Phi}'(x)| dx$$

Multivariate case: $\bar{\Phi}: \mathbb{R}^n \rightarrow \mathbb{R}^n$ $y = \bar{\Phi}(x)$

$$\int_{\mathbb{R}^n} g(y) dy = \int_{\mathbb{R}^n} g(\bar{\Phi}(x)) \cdot |\det(\partial_x \bar{\Phi})| dx$$

$$\bar{\Phi}(x+\Delta x) \approx \bar{\Phi}(x) + J_{\bar{\Phi}}(x) \cdot \Delta x, \quad J_{\bar{\Phi}}(x) = \partial_x \bar{\Phi}(x)$$

$dx \rightarrow$ ratio of volumes: $|\det J_{\bar{\Phi}}(x)|$

$$dy = |\det J_{\bar{\Phi}}(x)| dx, \quad \text{by } \Delta y \approx \bar{\Phi}'(x) \cdot \Delta x$$

$$\int_{\mathbb{R}^n} g(y) dy = \int_{\mathbb{R}^n} g(\bar{\Phi}(x)) \cdot |\det J_{\bar{\Phi}}(x)| dx, \text{ where}$$

$$\text{write: } \bar{\Phi}(x) = \begin{bmatrix} \bar{\Phi}_1(x_1, \dots, x_n) \\ \bar{\Phi}_2(x_1, \dots, x_n) \\ \vdots \\ \bar{\Phi}_n(x_1, \dots, x_n) \end{bmatrix}$$

$$J_{\bar{\Phi}}(x) = \frac{\partial \bar{\Phi}}{\partial x}(x) = \begin{bmatrix} \frac{\partial \bar{\Phi}_1}{\partial x_1} & \frac{\partial \bar{\Phi}_1}{\partial x_2} & \dots & \frac{\partial \bar{\Phi}_1}{\partial x_n} \\ \frac{\partial \bar{\Phi}_2}{\partial x_1} & \frac{\partial \bar{\Phi}_2}{\partial x_2} & \dots & \frac{\partial \bar{\Phi}_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial \bar{\Phi}_n}{\partial x_1} & \frac{\partial \bar{\Phi}_n}{\partial x_2} & \dots & \frac{\partial \bar{\Phi}_n}{\partial x_n} \end{bmatrix}$$

$$[J_{\bar{\Phi}}(x)]_{ij} = \frac{\partial \bar{\Phi}_i}{\partial x_j}(x)$$

We are ready to prove:

$$p_1(x_1) = p_0(\Psi^{-1}(x_1)) \cdot \left| \det \left(\frac{\partial \Psi^{-1}}{\partial x_1} \right) \right|. \quad (\text{B.1.1})$$

Proof:

$y = \Phi(x)$, we want a function p_y for any test function φ

$$\begin{aligned} \mathbb{E}[\varphi(y)] &= \int_{\mathbb{R}^d} \varphi(y) \cdot p_y(y) dy \\ &= \mathbb{E}[\varphi(\Phi(x))] = \int_{\mathbb{R}^d} \varphi(\Phi(x)) \cdot p_x(x) dx \end{aligned}$$

By Dirac delta:

$$\varphi(\Phi(x)) = \int_{\mathbb{R}^d} \varphi(y) \cdot \delta(y - \Phi(x)) dy : \text{Pick the point}$$

Now:

$$\int_{\mathbb{R}^d} \varphi(\Phi(x)) \cdot p_x(x) dx = \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} \varphi(y) \cdot \delta(y - \Phi(x)) dy \right) \cdot p_x(x) dx$$

$$\begin{aligned} \text{swap} &= \int \varphi(y) \left(\int \delta(y - \Phi(x)) \cdot p_x(x) dx \right) dy \\ &= \int_{\mathbb{R}^d} \varphi(y) \cdot p_y(y) dy \end{aligned}$$

$$\Rightarrow p_y(y) = \int_{\mathbb{R}^d} \delta(y - \Phi(x)) \cdot p_x(x) dx$$

Delta under a diffeomorphism:

$$\delta(y - \Phi(x)) = \frac{\delta(x - \Phi^{-1}(y))}{|\Phi'(x)|} \quad \text{with } x = \Phi^{-1}(y)$$

$$p_y(y) = \int_{\mathbb{R}^d} \delta(y - \Phi(x)) \cdot p_x(x) dx$$

$$= \int \delta(x - \Phi^{-1}(y)) \cdot |\det J(\Phi^{-1}(y))| \cdot p_x(x) dx$$

$$= |\det(\frac{\partial \Phi^{-1}}{\partial y})| \cdot \int \delta(x - \Phi^{-1}(y)) \cdot p_x(x) dx$$

$$= |\det(\frac{\partial \Phi^{-1}}{\partial y})| \cdot p_x(\Phi^{-1}(y))$$

$\nwarrow$ pick point $\Phi^{-1}(y)$

■

Back to original coordinates:

$$p_0(x_0) = p_1(\Phi(x_0)) \cdot |\det \partial_x \Phi(x_0)|$$

Multiple Bijections:

$$x_k = \Phi_k(x_{k-1}) : x_0 \xrightarrow{\Phi_1} x_1 \xrightarrow{\Phi_2} \dots \xrightarrow{\Phi_L} x_L$$

$$x_0 \sim p_0 : p_1, \dots, p_L \text{ for } x_1, \dots, x_L$$

$$p_k(x_k) = p_{k-1}(x_{k-1}) \cdot \left| \det \frac{\partial \Phi_k}{\partial x_{k-1}} \right|^{-1}$$

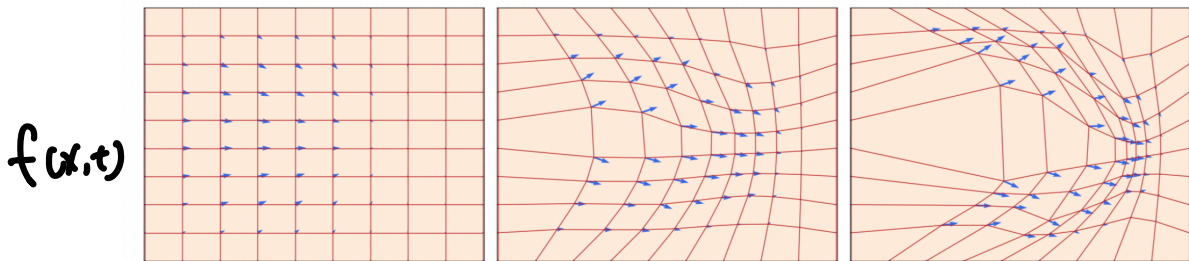
$$p_{x_L}(x_L) = p_{x_0}(x_0) \cdot \prod_{k=1}^L \left| \det \left(\frac{\partial \Phi_k}{\partial x_{k-1}} \right) \right|^{-1}$$

In log-density form:

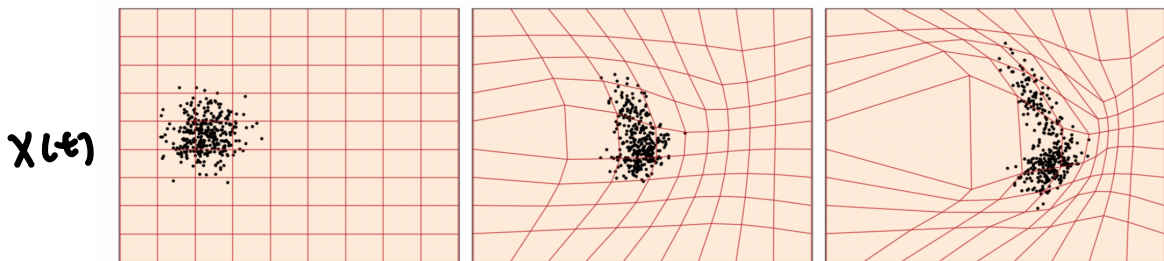
$$\log p_{x_L}(x_L) = \log p_{x_0}(x_0) - \sum_{k=1}^L \log \left| \det \left(\frac{\partial \Phi_k}{\partial x_{k-1}} \right) \right|.$$

$$p_{x_L}(x_L) = p_{x_0}(x_0) \cdot \prod_{k=1}^L \left| \det \left(\frac{\partial \Psi_k}{\partial x_{k-1}} \right) \right|^{-1}. \quad (\text{B.1.2})$$

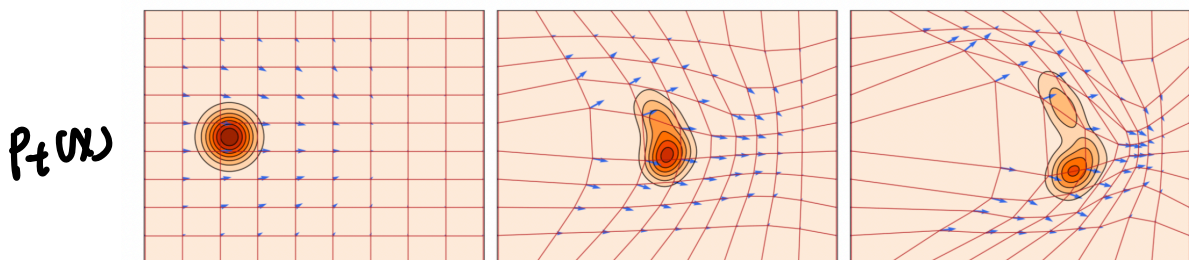
continuous flow governed by f : $\frac{dx(t)}{dt} = f(x(t), t)$. $x(0) = x_0 \sim p_0$



(a) Vector field illustrations. The arrows represent forces that would drag particles through space, deforming the underlying grid accordingly.



(b) Particle-cloud dynamics. A predefined vector field (interpreted as a force) generates a flow that transports particles from their initial state.



(c) Density evolution. As particles are advected by the vector field, the density contours deform accordingly, reflecting how the flow reshapes the underlying distribution.

Define: $X_{t+\Delta t} = \Phi(X_t) := X_t + \Delta t \cdot f(X_t, t)$, then:

$$\frac{dX(t)}{dt} = f(X(t), t), \quad X(0) = X_0 \sim p_0$$

$\Phi_{0 \rightarrow t} : \mathbb{R}^D \rightarrow \mathbb{R}^D$ a deterministic flow.

$$\Phi_{0 \rightarrow t}(X_0) = X_0 + \int_0^t f(X(\tau), \tau) d\tau$$

new density: $p_t = (\Phi_{0 \rightarrow t})_{\#} \cdot p_0$

$$p_t(x) = p_0(\Phi_{t \rightarrow 0}(x)) \cdot |\det \partial_x \Phi_{t \rightarrow 0}(x)|$$

$$= \int \delta(x - \Phi_{t \rightarrow 0}(x_0)) \cdot p_0(x_0) dx_0$$

Continuity Equation:

$$\frac{\partial}{\partial t} p_t(x) + \nabla \cdot (p_t(x) f(x, t)) = 0. \quad (\text{B.1.4})$$